What Counts as a Price?

Grain Prices and Market Integration in the Arsinoite Nome, First and Second Centuries CE

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Slides + PDF
ancientrome220.github.io/talks/triad-2026/

One account, different descriptions

P.Mich. II 127 · Tebtynis · selected wheat purchases, 45 CE

Limited variation

Rathbone
After the initial rise

Changing sequence

Bang
The rise and subsequent changes

Average level

Harper
A value for a longer series

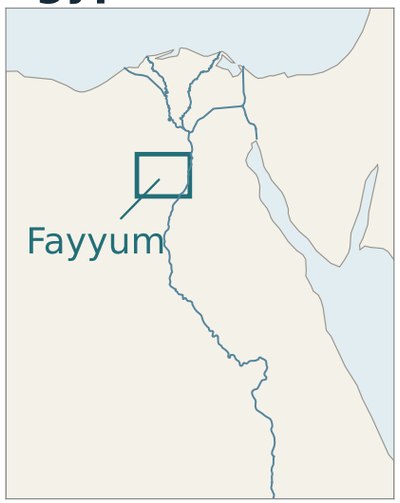
> What does each description preserve?

Rathbone 1997, p. 196; Bang 2011, pp. 171–172; Harper 2016, pp. 813–814, and accompanying database, sheet 2.

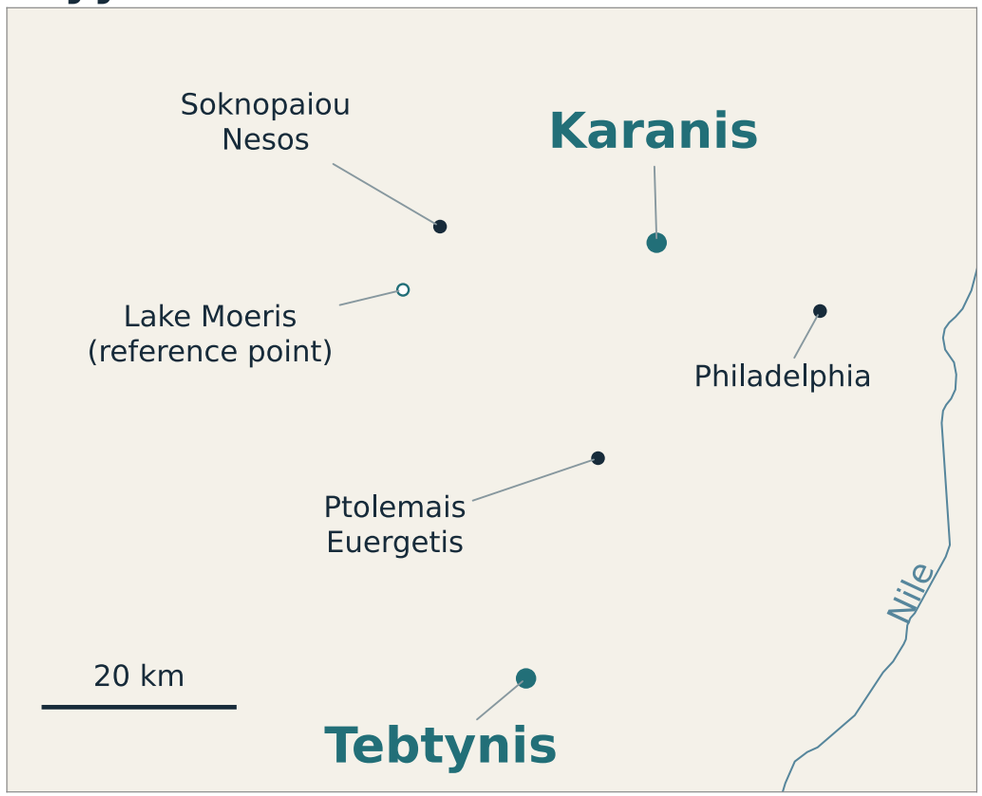
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Can we compare prices across localities?

Egypt



Fayyum / Arsinoite nome



Modern physical base; points locate places, not transactions.

Arsinoite nome / Fayyum
First and second centuries CE

A nome was an administrative district.

Exchange: transactions with variable terms.

Integration: linked supply and demand across places and time.

Our test: comparable wheat purchases in the same year?

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Erdkamp 2005, pp. 9–10; Temin 2013, pp. 7–8; manuscript §§1, 3.1, Fig. 1; map: Pleiades contributors and Natural Earth, consulted 9 November 2026. Modern base; neither ancient shorelines nor transaction venues reconstructed.

Define the amount and the observation

Reading an amount

Drachma: monetary unit
Artaba: capacity measure

Total payment ÷ quantity
= calculated rate

Counting evidence

Quantity + payment + stated rate
↓ One entry
↓ Several entries in one account

Amounts, entries, and accounts are different counting units.

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Manuscript §§3.1–3.2; Drexhage 1991, pp. 5–11; Rathbone 1997, pp. 186, 196, 240 n. 12. Shared unit names do not establish identical capacities.

The same payment buys different quantities

col. i, line 8

τιμῆ(ς) (πυροῦ) (ἀρταβῶν) α ∠ γ. (δραχμαὶ) η

price of wheat, 1 + ½ + ⅓ artabae: 8 drachmas

col. i, line 17

τιμῆ(ς) (πυροῦ) (ἀρτάβης) α (δεκάτου). (δραχμαὶ) η

price of wheat, 1 artaba and a tenth: 8 drachmas

col. i, line 8 · a priest

8 ÷ (11/6) ≈ **4.36**

drachmas per artaba, calculated

col. i, line 17 · Herodion

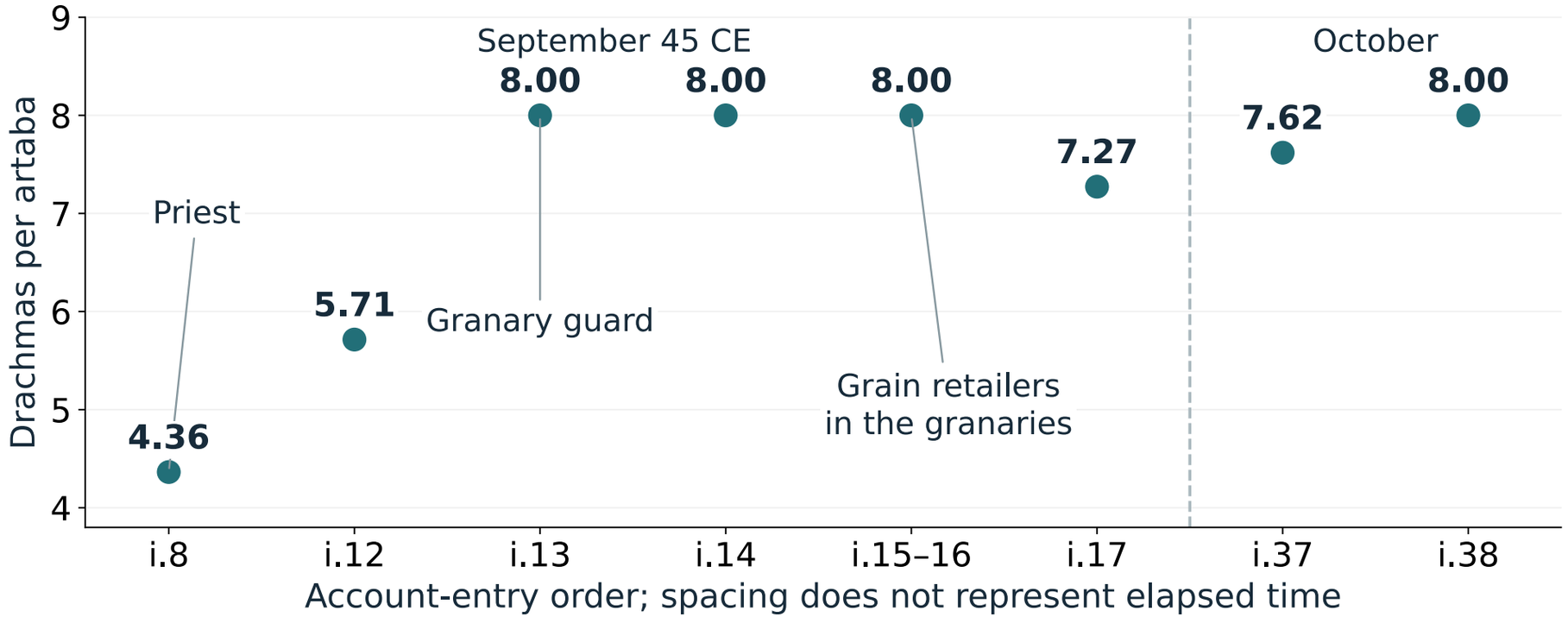
8 ÷ (11/10) ≈ **7.27**

drachmas per artaba, calculated

P.Mich. II 127, col. i, ll. 8, 17, electronic edition; manuscript §4.1, Table 2. Parentheses mark editorial expansions. Values rounded to two decimals; no new collation of the papyrus image.

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Eight purchases within one account



Rates rise from 4.36 to 8.00 within September, then stay between 7.27 and 8.00.

P.Mich. II 127, col. i, lines 8, 12–17, 37–38; manuscript Table 2. The whole account spans September 45–January 46; these eight selected entries span September–October 45.

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The disagreement concerns historical inference

Reading	Feature emphasised	Historical issue
Rathbone	Variation after the rise	How stable were local prices?
Bang	Rise, changes, and broad bands	What supports an integration inference?
Harper	An average price level	How does this account enter a longer series?

Rathbone 1997, pp. 196–198, 211–212; Bang 2011, pp. 171–172; Harper 2016, pp. 813–814, and database, sheet 2.

Slides

Aggregation changes the described pattern

Summary	September 45 CE						October		Drachmas per artaba
	4.36	5.71	8.00	8.00	8.00	7.27	7.62	8.00	
	i.8	i.12	i.13	i.14	i.15–16	i.17	i.37	i.38	
Average of all eight every entry counts once	●	●	●	●	●	●	●	●	7.12
Total paid ÷ total bought dot size = quantity bought	●	●	●	●	●	●	●	●	6.99
Repeated rate counted once three September 8.00s become one	●	●	●	●	●	●	●	●	6.83
Only after the initial rise first two entries left out; a range	○	○	●	●	●	●	●	●	7.27–8.00

> The third summary resembles Harper's published **6.83**; his exact entry mapping remains unresolved.

Manuscript §4.1; calculated with exact fractions. Harper database, sheet 2: average 6.83; separate plotting value 6.8. Recorded quantities have not been capacity-standardised.

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What would a cross-locality comparison require?

Comparable events

Wheat purchases or sales
The same event year

Comparable settings

Supported transaction localities
Compatible money and capacity bases

> Each condition needs support in the text, beyond catalogue provenance and document dates.

Manuscript §3.2, Table 1. “Usable” in the preliminary review means usable for within-account description; cross-locality comparison requires these additional conditions.

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Sale and provision record different events

SB VIII 9699 · Hermopolite account · February 79 CE

Sale wording · line 455

τιμῆ(ς) (πυροῦ) (ἀρταβῶν) δ
πεπραμένω(ν) εἰς δαπ(άνην)

“price of four artabae of wheat
sold for expenditure”

11

drachmas per artaba, calculated
4 artabae sold for 44 drachmas

Provision wording · lines 456–457

τιμῆ(ς) τῆς διατεταχθ(είσης)
αὐτῷ δι᾽ ἡμῶν πρὸς εὐθηνίαν
(πυροῦ ἀρτάβης) α

“price of the one artaba of wheat
assigned to him through us for provision”

8

drachmas for one artaba
Assigned for provision

Slides

Outside-region comparison. SB VIII 9699, lines 450–457; manuscript §3.2; Rathbone 1997, pp. 217, 221. Excerpts from the electronic edition; parentheses mark expansions. Line 454 states 11 for a separate entry. Compulsion, a specific discount, payment mechanism, and capacity equivalence remain unresolved.

From electronic texts to candidate price evidence

Workflow used for the TRIAD sample

Edited Greek texts

Electronic editions

→

Rule-based candidates

Grain, quantities, monetary expressions

→

LLM proposes relationships

Links quantities and amounts

→

Formal output checks

Fields, values, text-span bounds

↓

Historical source review — What does the recorded amount represent?

> Candidate relationships require historical interpretation.

Slides

Manuscript §3.3 and Appendix A.1; Fujimoto, Ogawa, and Ohmukai 2026, v12, §§3.2, 4.1–4.2. Formal checks assess output structure; historical readings require source review.

One entry can express two relationships

PSI V 543, line 7

Philadelphia • mid-third century BCE

Interlinear stated rate: 4 obols per artaba

ἡ ἀρ(τάβη) (τετρωβόλου)

κριθῶν (ἀρτάβη) α ∠

Quantity: 1½ artabae of barley

(γίνεται) (δραχμὴ) α

Total: 1 drachma

One entry, two relationships: quantity–total and unit quantity–rate

Quantity + total payment
1½ artabae + 1 drachma

Unit quantity + stated rate
1 artaba + 4 obols

Method illustration: Ptolemaic Philadelphia, 275–226 BCE, outside the Roman sample.

PSI V 543, l. 7; HGV 2165; manuscript Appendix A.2; Fujimoto, Ogawa, and Ohmukai 2026, v12, §4.2. Schematic from the electronic edition, not a facsimile. Both relationships describe one entry.

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Many extracted groups share documents

Candidate group: outputs sharing supporting text locations; competing readings may remain.

199

Candidate groups

all reviewed

156

Commodity entries

43 stated-rate and total pairs linked

54

Parent containers

from 69 electronic texts

One bank roll, three counting levels

82 groups → **14** electronic texts → **1** parent roll

Manuscript §§3.3, 4.2. Selection: recorded Arsinoite provenance, both date bounds present, interval overlapping 1–200 CE. It excludes records with missing date bounds or unrecognised provenance, and passages where the model returned no relationship. The counts are not independent transactions.

Slides

Screening defines the comparison sample

199 candidate groups, by review outcome

8	Usable for within-account wheat description
140	Contextual evidence
33	Unresolved readings or interpretation
18	Customs charges, rejected as grain prices

> All eight usable entries belong to **one account in 45 CE**.

Manuscript Table 3; preliminary source review. Contextual evidence includes administrative amounts, incomplete purchases, and barley excluded from the wheat comparison. Disposition is not extraction accuracy.

Slides

Other localities, no strict match

Document	Place and date	For a wheat-purchase comparison
P.Mich. II 127	Tebtynis · 45 CE	Eight usable entries, one account
SB XX 14576	Philadelphia · 43 and 46–47 CE	“Price of wheat, 6 drachmas”; no quantity; event date and purpose unresolved
P.Louvre I 23	Soknopaiou Nesos · 49 CE	Wheat-price payment for the seventh year; quantity and purpose unresolved
P.Cair.Goodsp. 30	Karanis · 192 CE	Five nominal subentries; private-sale classification unresolved

Manuscript §§3.3, 4.2–4.3, Fig. 3; preliminary source review. SB XX 14576, l. 34, and P.Louvre I 23, ll. 7–10: electronic editions, with HGV catalogue dates. Karanis dating: Rathbone 1997, p. 218. Provenance is not automatically transaction locality.

Slides

What the evidence supports

Within the account

Follow the purchase sequence
Select the post-rise interval
Group or weight the entries

Across localities

No strict contemporaneous match under the stated criteria

Integration remains unresolved.

> Missing matches establish neither integration nor isolation.

Manuscript §§4.3–5. The result is conditional on documentary survival, sample selection, source interpretation, event dates, and compatible measures.

Slides

Two judgments shape a price comparison

1 Identify what the amount records.

2 Decide what the summary preserves.

Discussion: which additional documents would make a cross-locality comparison possible?

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Thank you · ありがとうございます · 감사합니다 · 谢谢大家

Manuscript §6. Preliminary source analysis reviewed and accepted by the author; unresolved readings and comparability limits remain. AI assistance and provenance are documented in the final backup slide.

References: scholarship

Bang, Peter F. 2011. *The Roman Bazaar: A Comparative Study of Trade and Markets in a Tributary Empire*. Cambridge University Press. Especially pp. 171–173.

Drexhage, Hans-Joachim. 1991. *Preise, Mieten/Pachten, Kosten und Löhne im römischen Ägypten bis zum Regierungsantritt Diokletians*. Scripta Mercaturae. Especially pp. 5–11, 22.

Erdkamp, Paul. 2005. *The Grain Market in the Roman Empire: A Social, Political and Economic Study*. Cambridge University Press. Especially pp. 9–10.

Harper, Kyle. 2016. "People, Plagues, and Prices in the Roman World: The Evidence from Egypt." *Journal of Economic History* 76 (3): 803–839. doi:10.1017/S0022050716000826.

Rathbone, Dominic W. 1997. "Prices and Price Formation in Roman Egypt." In *Économie antique: Prix et formation des prix dans les économies antiques*, edited by Jean Andreau, Pierre Briant, and Raymond Descat, 183–244. Musée archéologique Départemental.

Temin, Peter. 2013. *The Roman Market Economy*. Princeton University Press. Especially pp. 7–8.

Slides

References: editions, method, and data

Electronic editions: Duke Databank of Documentary Papyri (DDbDP), through Papyri.info. P.Mich. II 127, col. i, lines 8, 12–17, 37–38; SB VIII 9699, ll. 450–457; PSI V 543, l. 7; P.Cair.Goodsp. 30, xiii.11–13, xv.24, xx.22–23; SB XX 14576, l. 34; P.Louvre I 23, ll. 7–10; P.Oxy. XLI 2958, ll. 4–13, and 2960, ll. 18–25. No new image collations.

Metadata: Heidelberger Gesamtverzeichnis der griechischen Papyrusurkunden Ägyptens (HGV); Trismegistos (TM). HGV 11972 and 2165 for the principal account and PSI example.

Harper, Kyle. 2016, database. *Database of Prices, Wages, and Rents for Roman Egypt from the First through Seventh Centuries AD*. DARMC Scholarly Data Series 2016-5. Sheet 2, consulted 23 October 2026.

Fujimoto, Shunya, Jun Ogawa, and Ikki Ohmukai. 2026. “Extracting Grain Price Information from Papyri while Preserving Editorial Structure: Rule-Based Candidate Enumeration and Selection by Large Language Models.” Jinmonkon submitted manuscript, v12, 31 October 2026. Japanese text with English title and abstract.

Map: Pleiades contributors, *Pleiades Data for GIS* (CC BY 3.0); Natural Earth, *Physical Vector Data* (public domain). Consulted 9 November 2026.

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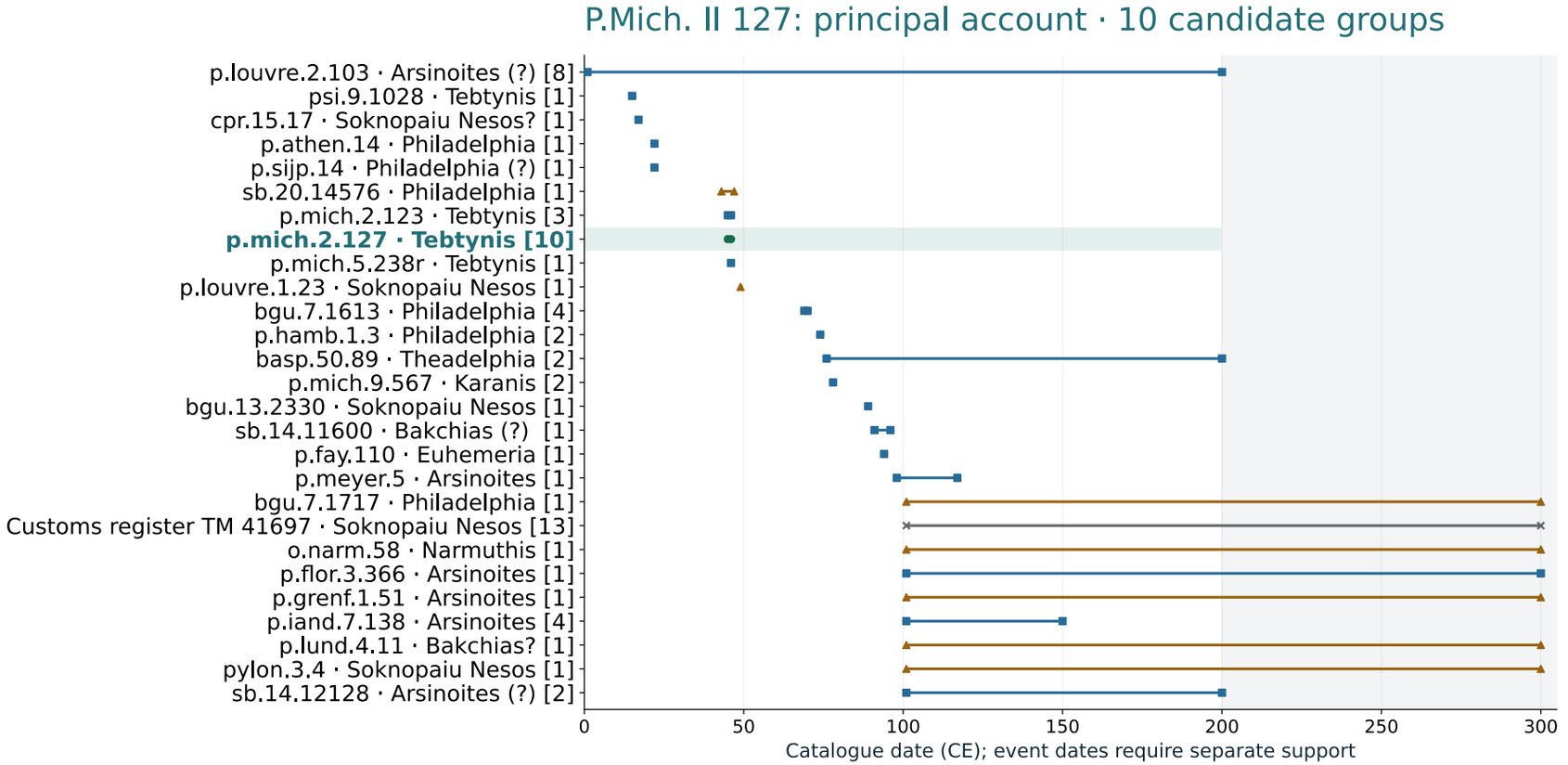
Backup: the eight selected wheat purchases

Col. i line	Period, 45 CE	Seller / description	Artabae	Drachmas	Rate
8	September	A priest	1 + ½ + ⅓	8	4.36
12	September	Orsenouphis	3 + ½	20	5.71
13	September	A granary guard	2	16	8.00
14	September	Aphrodisios	1	8	8.00
15–16	September	Grain retailers in the granaries	3	24	8.00
17	September	Herodion	1 + 1/10	8	7.27
37	October	Aphrodisios	2 + 1/10	16	7.62
38	October	Not named in this entry	1 + ½	12	8.00

P.Mich. II 127, electronic edition; manuscript Table 2. Rates: nominal drachmas/artaba, rounded to two decimals. Seller descriptions are working translations; fractions follow the expanded edition.

Slides

Backup: documentary coverage, rows 1–27



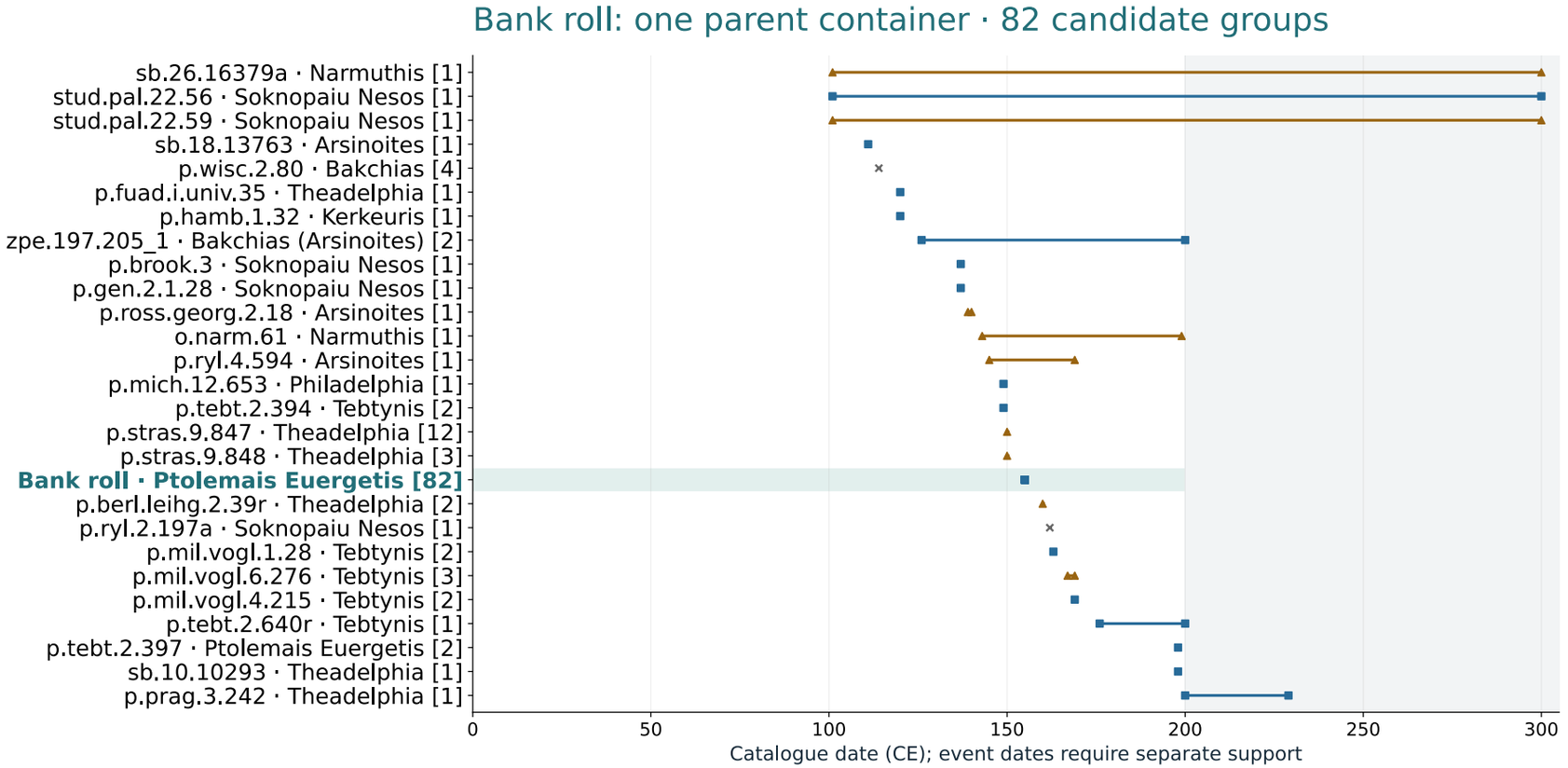
● Usable present ■ Contextual ▲ Unresolved × Customs

Colour/shape shows the most inclusive disposition in each container; [n] counts groups.

Slides

Manuscript Fig. 3. Catalogue provenance and date intervals; not transaction venues or event dates. Together the two panels preserve all 54 rows and 60 intervals.

Backup: documentary coverage, rows 28–54



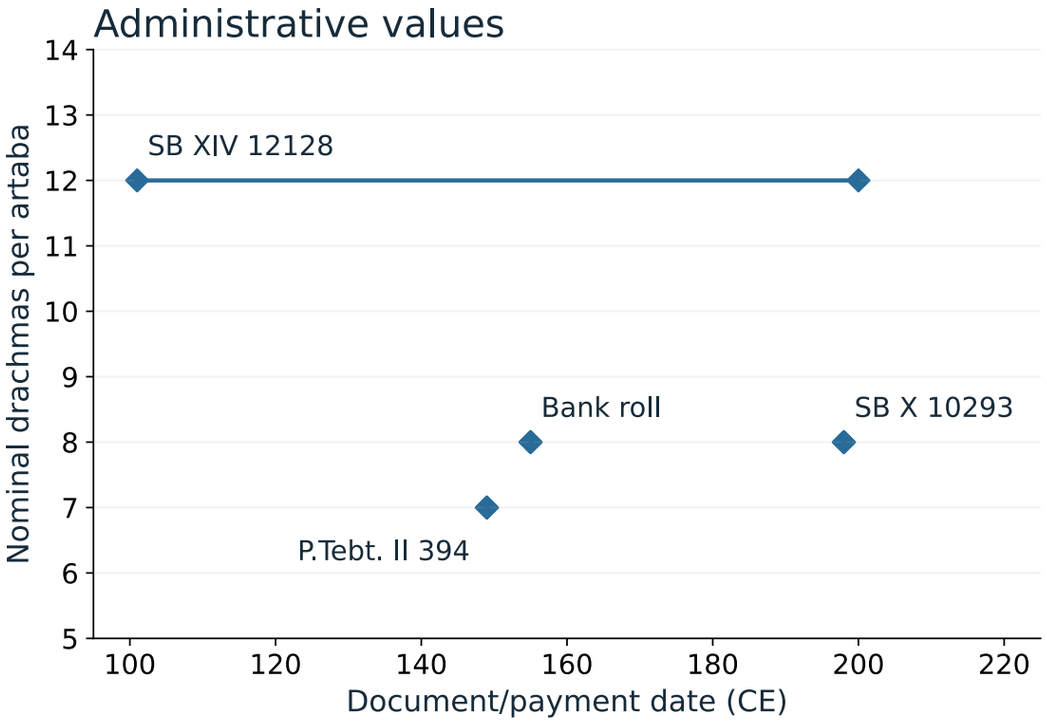
● Usable present ■ Contextual ▲ Unresolved × Customs

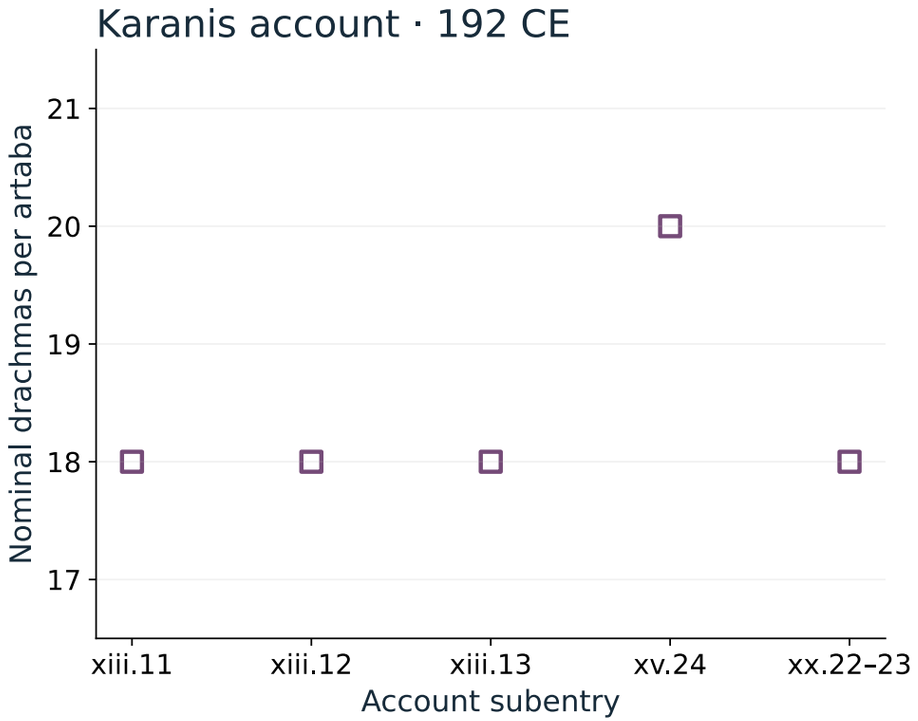
Colour/shape shows the most inclusive disposition in each container; [n] counts groups.

Slides

Manuscript Fig. 3. Broad intervals remain intervals; the shaded zone lies beyond 200 CE. Place labels are abbreviated for display.

Backup: keep documentary contexts separate





Different vertical scales. Original measures retained; no regional series or common capacity conversion.
Karanis: January–February 192 CE; private-sale classification unresolved.

Slides

Manuscript Fig. 4, panels B–C. Rathbone 1997, p. 218, for Karanis dating. Bank-roll marker summarises repeated stated rates; administrative dates are document/payment dates.

Backup: extraction agreement and its limits

Model	Words/side	Span precision	Span recall	Span F1	Complete F1
GPT-6.1 Sol	25	21/45	21/36	0.519	0.395
Claude Fable 5.1	25	22/41	22/36	0.571	0.442
Gemini 3.8 Flash	25	21/38	21/36	0.568	0.405
GPT-6.1 Sol	80	20/50	20/36	0.465	0.372
Claude Fable 5.1	80	20/41	20/36	0.519	0.468
Gemini 3.8 Flash	80	21/34	21/36	0.600	0.429

150 mentions / 142 documents; 36 reference pairs. Development overlap, one reviewer with AI assistance, one run per condition. Exact spans first; complete matching also requires values, units, commodity, states, and rate basis.

Manuscript Appendix A, Table A1; Fujimoto, Ogawa, and Ohmukai 2026, v12, Table 4. 28–29 October 2026; target-grain-prompt-v2. Excerpts count words, not tokens. Independent-sample performance and general model rankings remain untested.

Slides

Backup: a petition does not date price formation

Document	Petition date	Crop named
P.Oxy. XLI 2958	December 99 CE	Trajan’s second year
P.Oxy. XLI 2960	January 100 CE	Nerva’s first year

> Crop year, petition date, price formation, and payment are distinct events.

Outside-region examples from the Oxyrhynchite nome.

Slides

P.Oxy. XLI 2958, ll. 4–13; 2960, ll. 18–25; Rathbone 1997, p. 221; manuscript §3.2. Preserve the crop years in their regnal form; eventual receipt of payment may remain uncertain.

Backup: the compilation is a separate route

Cross-check

26 price rows
19 source entries
7 electronic texts absent from the extracted sample

Selection limits

Missing date bounds
Unrecognised provenance
No relationship returned

P.Lund IV 11: Rathbone favours third-century dates, but also mentions 170 CE. Retain as a separate dating sensitivity case.
P.Cair.Goodsp. 30: missing upper date bound excludes it from the extracted sample.

Manuscript §§3.3, 4.2; compilation cross-check; Rathbone 1997, pp. 217–219, 221–222. Broader date intervals remain intervals. Compilation rows are not added to the 199 candidate groups.

Slides

Backup: analysis provenance and AI assistance

Extraction: language models and conditions reported in Appendix A; Gemini 3.8 Flash with 80 words per side supplied the full candidate collection.

Preparation: OpenAI Codex assisted source classification, documentary links, analysis code, figures, drafting, and checking. Grammarly assisted manuscript language revision. Codex prepared this deck and script.

Author review: the author reviewed the preliminary analysis and accepted it without changes on 9 November 2026.

Evidence status: the candidate-level record still marks each item as AI-assisted source review, not author adjudication. Unresolved readings and limits of comparability remain. Responsibility rests with the author.

Manuscript “Use of AI tools”; project README and review record. Author acceptance is not a new item-by-item adjudication or an independent source verification.

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